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main.c

Run

```
1 #include <stdio.h>
2 int main() {
3     int A[10][10], B[10][10], C[10][10];
4     int r1, c1, r2, c2;
5     int i, j, k;
6     printf("Enter rows and columns of first matrix: ");
7     scanf("%d %d", &r1, &c1);
8     printf("Enter rows and columns of second matrix: ");
9     scanf("%d %d", &r2, &c2);
10    if (c1 != r2) {
11        printf("Matrix multiplication is not possible.\n");
12        return 0;
13    }
14    printf("Enter elements of first matrix:\n");
15    for (i = 0; i < r1; i++) {
16        for (j = 0; j < c1; j++) {
17            scanf("%d", &A[i][j]);
18        }
19    }
20    printf("Enter elements of second matrix:\n");
21    for (i = 0; i < r2; i++) {
```

Output

Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
1 2
2 3
Enter elements of second matrix:
3 4

4 5
Product matrix:
11 14
18 23

=== Code Execution Successful ===

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main.cRun

```
2 int main() {
21 for (i = 0; i < r2; i++) {
22     for (j = 0; j < c2; j++) {
23         scanf("%d", &B[i][j]);
24     }
25 }
26 for (i = 0; i < r1; i++) {
27     for (j = 0; j < c2; j++) {
28         C[i][j] = 0;
29         for (k = 0; k < c1; k++) {
30             C[i][j] += A[i][k] * B[k][j];
31         }
32     }
33 }
34 printf("Product matrix:\n");
35 for (i = 0; i < r1; i++) {
36     for (j = 0; j < c2; j++) {
37         printf("%d\t", C[i][j]);
38     }
39     printf("\n");
40 }
```

Output

Enter rows and columns of first matrix: 2 2
Enter rows and columns of second matrix: 2 2
Enter elements of first matrix:
1 2
2 3
Enter elements of second matrix:
3 4
4 5
Product matrix:
11 14
18 23

=== Code Execution Successful ===